

AppStat 25/26 Problem Set

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In the following, all fits are done using `iMinuit`. Furthermore, for all parts where random number generation is used, the seed is fixed to 1, for reproducibility.

Problem I

1.1

This problem is solved by simulating 1 million days. Each day a dice-roll is simulated, and if a 6, 120 rolls are performed. The amount of sixes obtained from the 120 rolls follow a binomial distribution with $N = 120$, $p=1/6$. The amount of push-ups is obtained using numpy's binomial function. Finding the days with no push-ups and the days between, we get the following distribution::

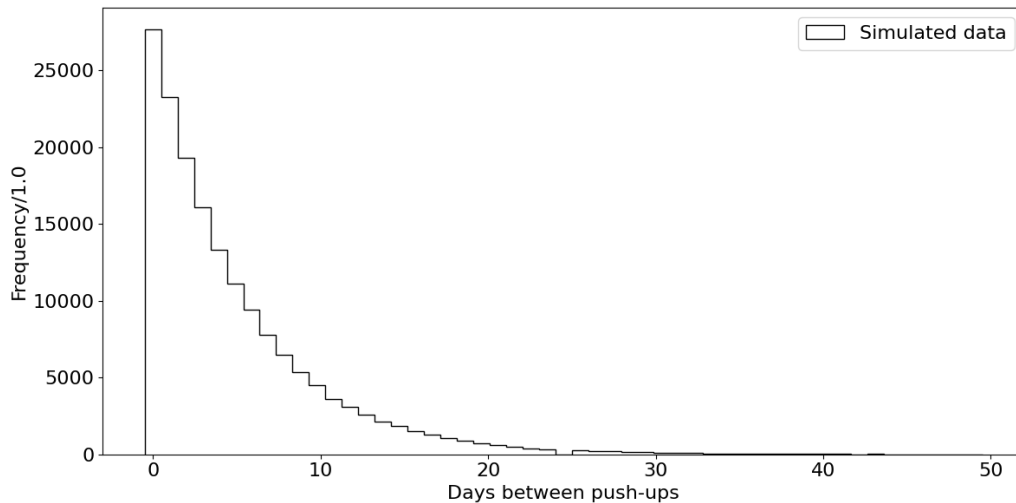


Figure 1: Histogram showcasing the distribution of days between push-ups.

From fig. 1 it is clear that it follows a geometric distribution. This makes sense, since it is determined by Bernoulli trials, where only one success is needed.(ADD MORE)

To obtain the mean 10 day push-up number, we split the results into arrays of length 10, and find the sum for each. The standard deviation is $\sigma_{10} = 24.21$. The mean is (33.30 ± 0.08) push-ups, with the error found using $\sigma_{10}/\sqrt{N}$, N for amount of 10-length entries. We found a median of 31.0.

1.2

We assume the Djoser pyramid is NS-aligned if its sides of the square base is NS-aligned. The problem is symmetric: if any pair of the sides are NS-aligned, the pyramid is NS-aligned. Needing only one pair of the sides, the pyramid has a 45-degree freedom to turn both left and right. Needing the pyramid to be NS-aligned by coincidence within 3 degrees, the probability that it is NS-aligned by coincidence is given as $6/90 = 1/15 \approx 6.67\%$.

Problem II

2.1

This problem is solved using error propagation, since the error on all the values are given. We assume the hydrogen ratio stems partly from the proto-solar system and partly from comets only. Thus some fraction, f , stems from the former, whilst the rest, $1 - f$, stems from the latter. We thus have the following equation:

$$r_{\otimes} = r_{\odot}f + r_C(1 - f) \quad (1)$$

yielding,

$$f = \frac{r_{\otimes} - r_C}{r_{\odot} - r_C} \quad (2)$$

Using Sympy, we can find an expression for the error on f . The error on $1 - f$ is the same, since the constant doesn't matter w.r.t. error propagation.

The expression is:

$$\sigma_f = \sqrt{\sigma_C^2 \left(\frac{r_{\otimes} - r_C}{(r_{\odot} - r_C)^2} - \frac{1}{r_{\odot} - r_C} \right)^2 + \sigma_{\otimes}^2 \frac{1}{(r_{\odot} - r_C)^2} + \sigma_{\odot}^2 \frac{(r_{\otimes} - r_C)^2}{(r_{\odot} - r_C)^4}} \quad (3)$$

We get $f_{\odot} = (0.56 \pm 0.03)$ and $f_C = (0.44 \pm 0.03)$.

2.2

The values for N , Δt and $\sigma_{\Delta t}$ are given. The uncertainty on N is assumed Poisson, giving $\sigma_N = \sqrt{N} = 44$.

The rate is calculated using the formula. The error is calculated using error propagation on the formula, again using sympy. We obtain:

$$\sigma_r = \sqrt{\left(\frac{N}{\Delta t^2} \sigma_{\Delta t} \right)^2 + \left(\frac{1}{\Delta t} \sigma_N \right)^2} \quad (4)$$

From this we obtain $r = (20.0 \pm 0.9)\text{counts/s}$.

To find what Δt should be for the relative error on r to be 2.5%, we assume the above found value of the rate is a good estimate for the actual rate. We then scan over values of Δt and calculate N . Using the function used above, we calculate new values of r and σ_r . Once we've reached $\sigma_r/r = 0.025$, we stop.

Doing this we reach a value of $\Delta t = 193.3\text{s}$.

2.3

2.3.1

For this part, the uncertainties given in the data-sheet are neglected. At first we manually sort the measurements into four groups each corresponding to measurements of one of the pylons, using the left-right cut-off limits showcased below. For each group, taking the measurements as repeated measurements of the same quantity, assuming a Gaussian distribution, the mean values and their errors are found using error on the mean. This is summarized in the following table:

Pylon no.	L/R limits	μ	σ (error)
1. Pylon	0/60	50.1 m	0.7 m
2. Pylon	60/120	91.1 m	1.0 m
3. Pylon	120/150	132.0 m	0.4 m
4. Pylon	150/none	172.1 m	0.5 m

Table 1: Table showcasing Left-Right value limits, mean pylon positions and corresponding errors.

To cross check, we fit each pylon-peak with a Gaussian function in an unbinned LLH fit. An unbinned LLH fit is appropriate since we are in the relatively low statistics regime with unbinned data.

First, we filter out any outliers using Chauvenets criterion by using the above mean-values and standard deviations. Using the accepted data, we get the following unbinned LLH fit:

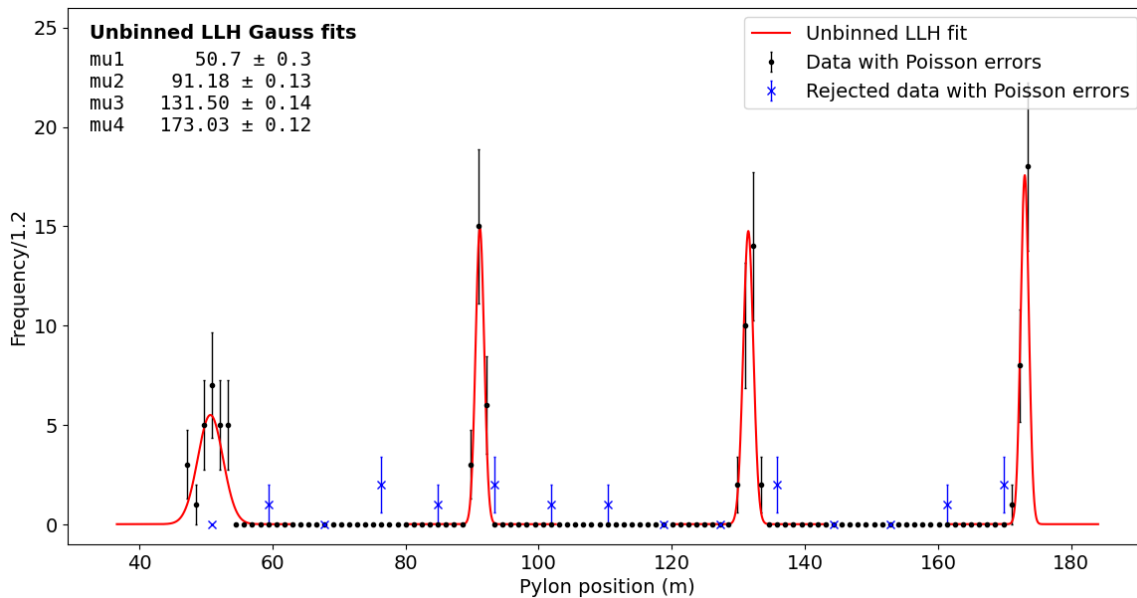


Figure 2: Unbinned LLH of pylon data. On the figure the accepted and rejected data with Poisson errors are shown as black dots and blue x's, respectively. The red lines are the unbinned LLH fit results. The found mean-values with uncertainties are drawn on top.

On the figure, we see the accepted and rejected data and the fit-results.

Combining the mean-values found in table 1 with the ones found in fig. 2 in a weighted mean, we get:

Pylon no.	Rep. meas.	Unbinned LLH fit	Weighted mean	z-value	p
1. Pylon	(50.1 ± 0.7) m	(50.7 ± 0.3) m	(50.6 ± 0.3) m	0.75	0.56
2. Pylon	(91.1 ± 1.0) m	(91.18 ± 0.13) m	(91.18 ± 0.13) m	0.03	0.98
3. Pylon	(132 ± 0.4) m	(131.50 ± 0.14) m	(131.57 ± 0.13) m	-1.25	0.21
4. Pylon	(172.1 ± 0.5) m	(173.03 ± 0.12) m	(172.99 ± 0.12) m	1.65	0.10

Table 2: Combined pylon position measurements without uncertainties in a weighted mean with z- and p-values displayed.

To see whether the measurements agree, a z-test was performed. As can be seen their p-values are all in the midrange, far away from any reasonable significant level. They must then agree.

2.3.2

Taking the given uncertainties into consideration, we first need to clean the dataset by removing any uncertainties below 0 - we assume this to mean that the data points are invalid.

We then assume that Using the filtered date, for each data, i.e. each pylon-measurements, we calculate the z-values and accept only data-points with $z < 2$, i.e. less than 2σ away, essentially doing a pull distribution. This is appropriate since we have assumed the measurements are distributed according to a Gaussian distribution.

The mean pylon positions with uncertainties are then obtained from a weighted mean, yielding:

Pylon no.	Weighted mean
1. Pylon	(50.6 ± 0.6) m
2. Pylon	(91.0 ± 0.4) m
3. Pylon	(132.1 ± 0.3) m
4. Pylon	(172.7 ± 0.4) m

Table 3: Pylon positions from weighted means taking uncertainties into account, using only valid data and only data with $z < 2$.

2.3.3

Combining the measurements in a weighted mean and calculating z-values, we get:

Pylon no.	Final weighted mean	z-values	p
1. Pylon	(50.6 ± 0.3) m	0.01	0.99
2. Pylon	(91.16 ± 0.12) m	0.44	0.66
3. Pylon	(131.65 ± 0.12) m	-1.55	0.12
4. Pylon	(172.96 ± 0.11) m	0.70	0.48

Table 4: Combined weighted mean of pylon positions found without and including uncertainties. The found z- and p-values are displayed.

The measurements agree with a significant level of 5%, but the first pylon position measurements does not survive a level of 1%.

2.3.4

To test if they are equidistant, we use the final weighted mean values and get the following results:

Pylons	Distance
1-2	(40.6 ± 0.3) m
2-3	(40.5 ± 0.2) m
3-4	(41.3 ± 0.2) m

Table 5: Distances between pylon with uncertainties.

As can be seen in table 5, pylons 1 through 3 are equidistant, but 3 and 4 are significantly more apart. The errors are found by adding the errors in quadrature and taking the square-root.

Problem III

3.1

This problem is solved using Monte-Carlo simulation. First we create a box around each circle and simulating random values inside the box. Then we accept points which obey:

$$||\mathbf{x}||_2 \leq r \quad (5)$$

i.e. all the points whose 2-norm is less than or equal to the radius of each circle.

The points inside each circle are then compared to each other and those who are equal are assumed to be the overlapping points.

For the 2d case with 1 million points we get:

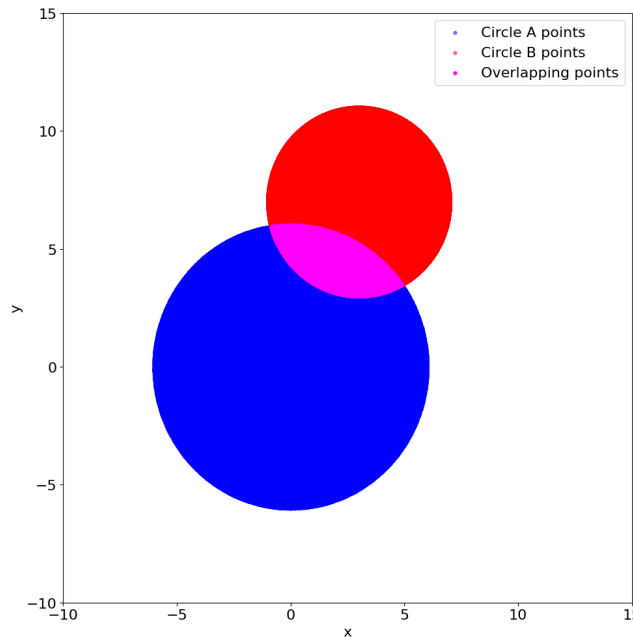


Figure 3: MC simulation of points inside two circles. The blue and red dots denote points inside circle A and B, respectively. The magenta points are the overlapping points.

The fraction of A which overlap B is 0.091 and the fraction of B which overlap with A is 0.204. The fractions are found by dividing the overlapping points with the amount of points from the parent circle.

This is generalized to the 4d case. For 1 million points, we get a the fraction of A which overlaps with B is 0.015 and the fraction of B which overlaps with A is 0.075.

The results are summed up in the table below:

Dim	$A \cap B / A$	$B \cap A / B$
2d	0.091	0.204
4d	0.014	0.074

Table 6: Fraction of A and B in each other, respectively, for the 2 and 4 dimensional cases.

3.2

3.2.1

Simulating 50000 points of $x, y, z \in [-1, 1]$ and plotting the radius using the 2-norm, we get:

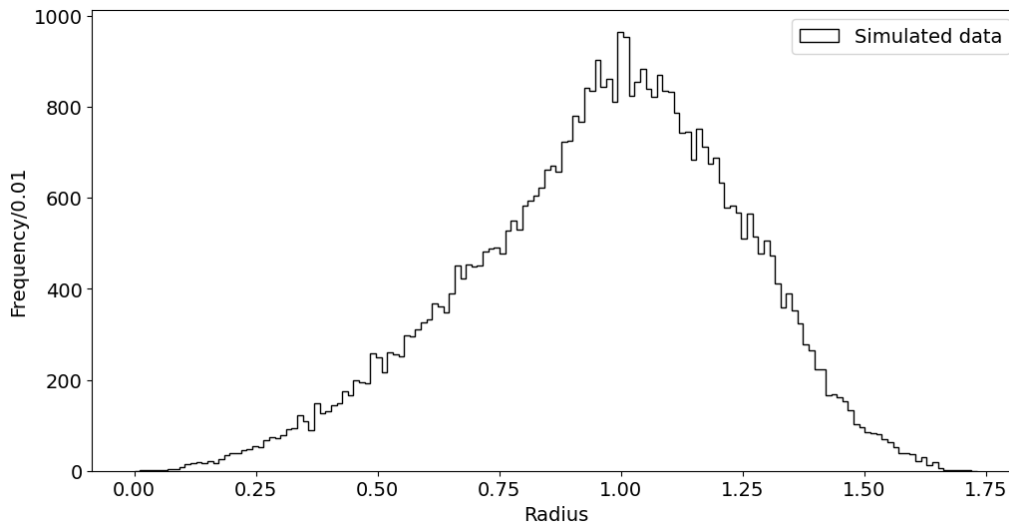


Figure 4: Simulated distribution of the radius from the explosion.

As can be seen on fig. 4, it is most frequent for the radius to be around 1.

3.2.2

We first simulate data according to the constraints that $z > 0$. Then we transform to spherical coordinates and force the constraint that $r < 1$.

We first get the θ distribution:

As can be seen on fig. 5, it is most likely to obtain large θ -values. This makes sense under the constraints, because they essentially limit the explosion to happen in the upper half of a sphere. Slicing the circle perpendicular to the z -axis, we see that the area is largest for large θ -values. Under assumption that the coordinate system is right handed.

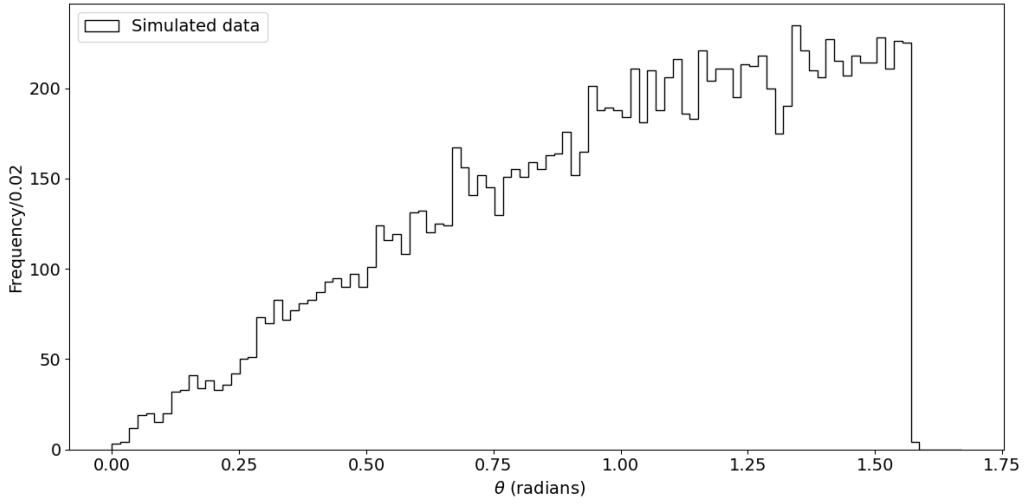


Figure 5: Simulated θ distribution under $z > 0$ and $r < 1$ constraints.

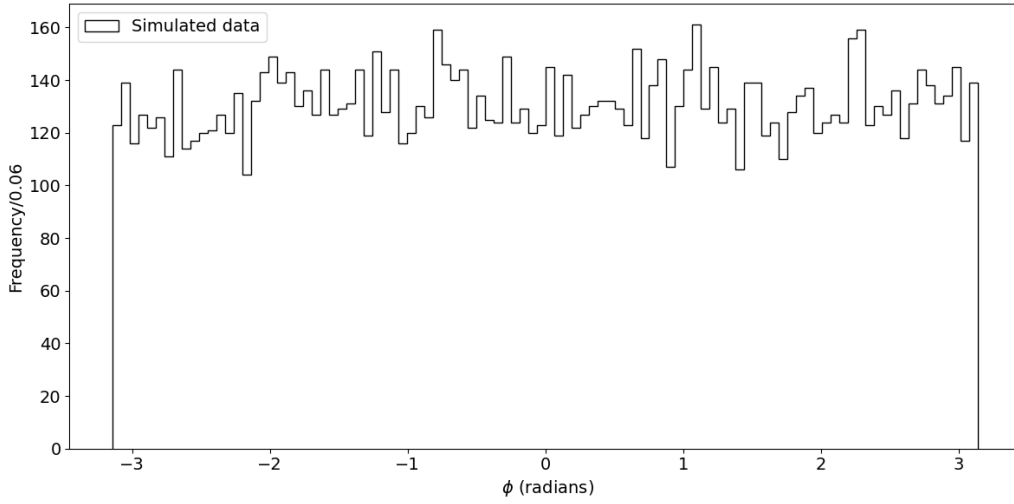


Figure 6: Simulated ϕ distribution under $z > 0$ and $r < 1$ constraints.

The ϕ distribution can be seen on fig. 6:

As can be seen in fig. 6, it ϕ approximately uniformly distributed. This makes sense, since the constraints do not affect ϕ .

3.2.3

To produce random velocities according to $f(v) = (v/v_0)^2 \exp(-v/v_0)$, we would use the Monto-Carlo method. We would first determine the interval over which this function is valid. Here we would assume $v \geq 0$ and $v \leq c$ since it is a velocity. Then we would normalize $f(v)$ by integrating it. Taking $c \approx \infty$ w.r.t. normal velocities, we get:

$$F(v) = (1/v_0)^2 \int_0^\infty dv (v^2 \exp(-v/v_0)) = (1/v_0)^2 2v_0^3 = 2v_0 \quad (6)$$

Thus, normalized we have:

$$f(v) = \frac{1}{2v_0} (v/v_0)^2 \exp(-v/v_0) \quad (7)$$

I would then use accept/reject for a range of points on eq. (7).

3.2.4

To find the distribution of $m(r)$, we'll use the simulated θ -values. We then only need to simulate values of v according to eq. (7). We have ensured that it is normalized.

Using accept/reject on eq. (7), with the constraint that the amount of v -values must equal the amount of θ -values, we get the following distribution of v :

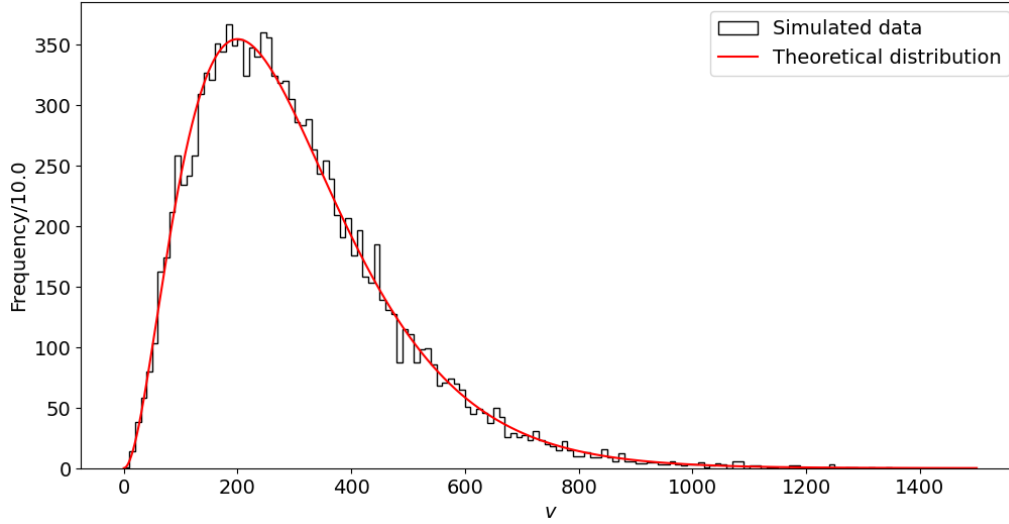


Figure 7: Simulated values of v according to eq. (7) with theoretical distribution on top.

The theoretical distribution is drawn on top. The simulated values and theoretical distribution match well.

Combining the simulated v - and θ -values, using:

$$r(v) = \sin(\theta)v^2/g \quad (8)$$

we get the following material distribution:

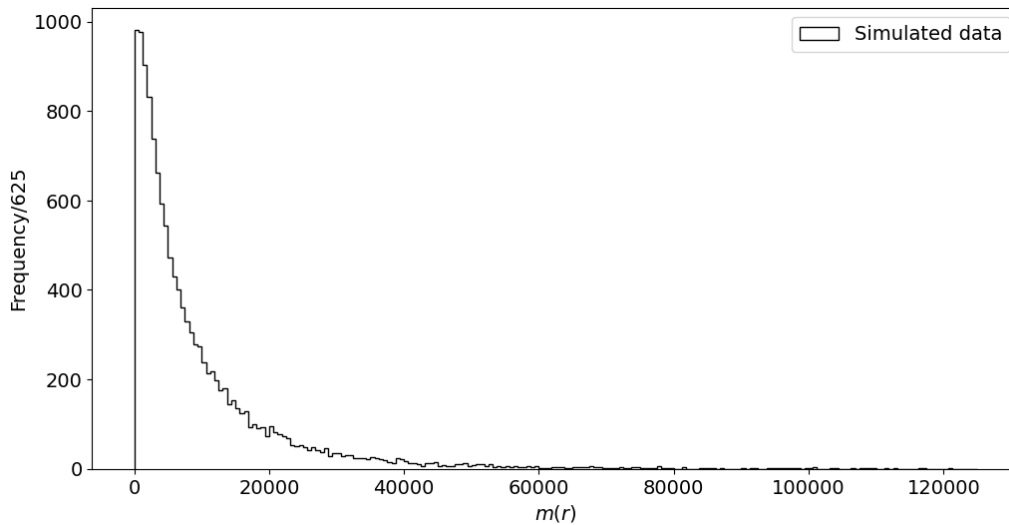


Figure 8: Simulated material distribution, $m(r)$ according to eq. (8).

Problem IV

4.1

4.1.1 & 4.1.2

This is simulated in the following way. For the box containing only real dice, for N rolls, we randomly simulate 25 integers between 1 and 6 and store them, using numpy's `randint` function. For the box with one fake die, we simulate for N rolls 24 random integers between 1 and 6, and then append N of the same random digit between 1 and 6 to the list. The function has an incorporated mixer, which permutes the order of the non-fake and fake runs.

To make a statistical test, we may setup the following null-hypothesis:

H_0 : The frequency of the digits will distribute themselves in a uniform manner.

With the alternative hypothesis being that they will not distribute themselves uniformly, we set a significant level of 1%.

Calculating Pearson's χ^2 according to:

$$\chi^2 = \sum_i \frac{(O_i - E_i)^2}{E_i} \quad (9)$$

and the corresponding p-value, we can test whether our runs contain a fake die or not. Simulating two runs, one of which contains one fake die, we get the following frequency distribution for $N = 200$ for the first run:

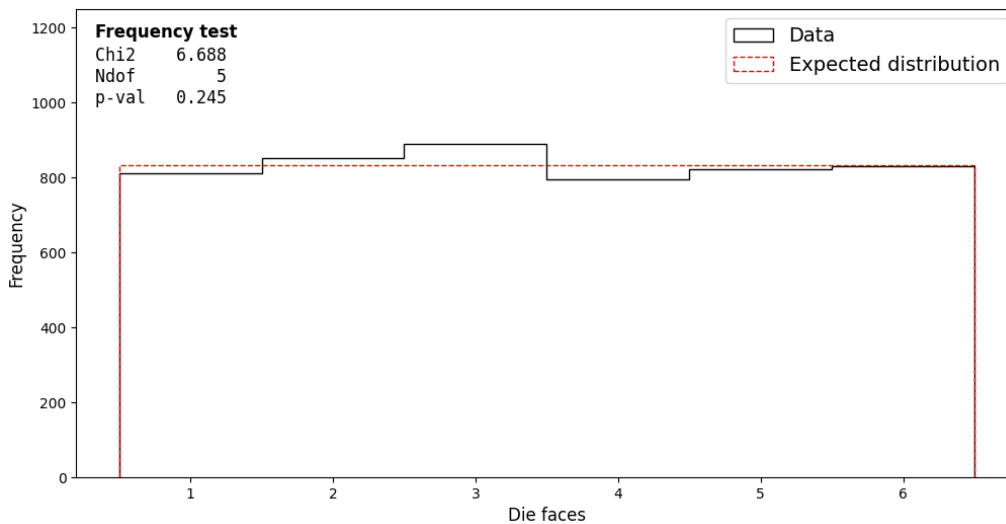


Figure 9: Die face distribution for run 1 with χ^2 , Ndof and p-value displayed.

Getting a p-value of $p \approx 0.25$, the found distribution is not significant, and we accept H_0 for run 1.

For run 2, we get:

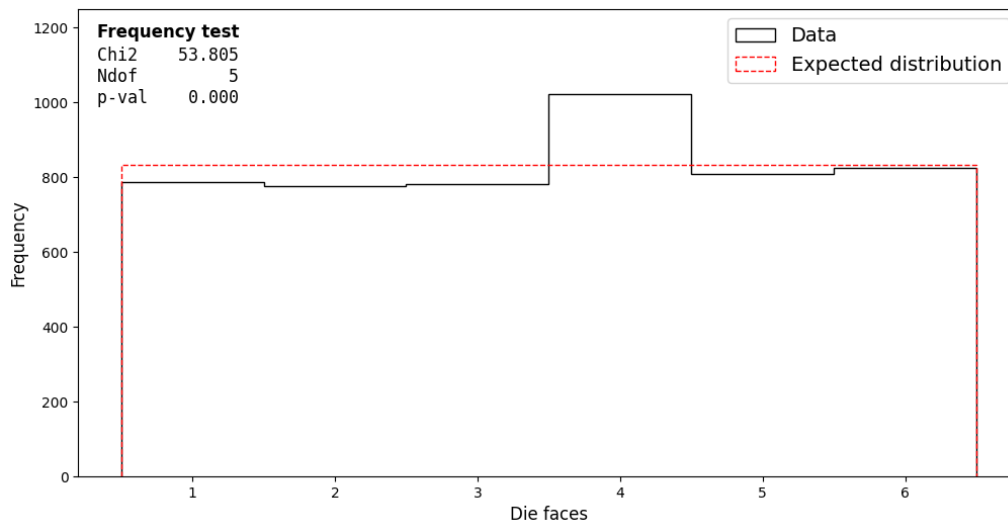


Figure 10: Die face distribution of run 2 with χ^2 , Ndof and p-value displayed.

Getting a p-value a lot lower than our significant level of 1%, we reject H_0 . Looking at fig. 10, the fake die contains only 4 as a die face.

4.1.3

To test how many rolls, N , needed before we can tell whether our box contains a fake die, we'll use the same approach as above with the same significant level of 1%.

Iterating over N rolls and calculating p -values, and stopping once we've hit the level of significance, and simulating it 50000 times, we get the following distribution of N rolls needed before $p < 1\%$:

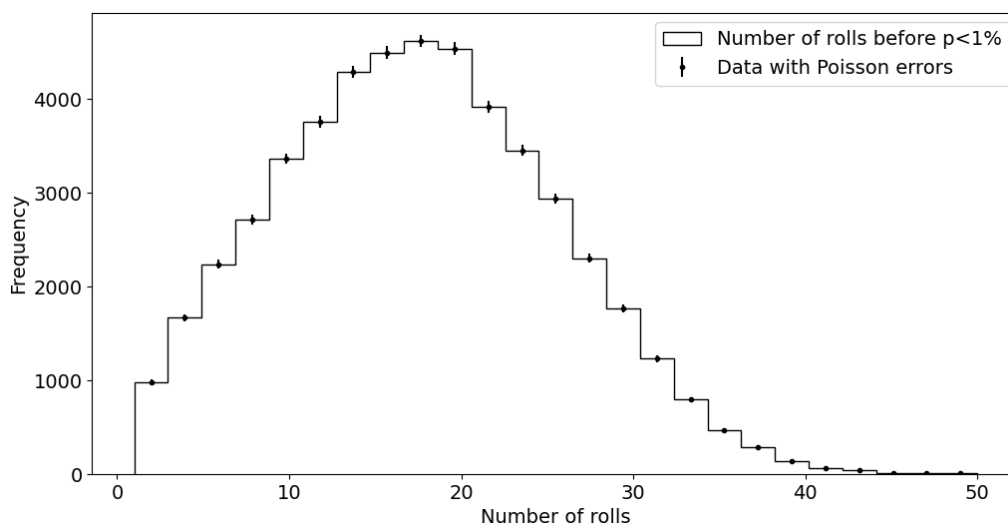


Figure 11: Distribution of number of rolls before $p < 1\%$

Doing a χ^2 -fit on top using a Gaussian function, we get the following:

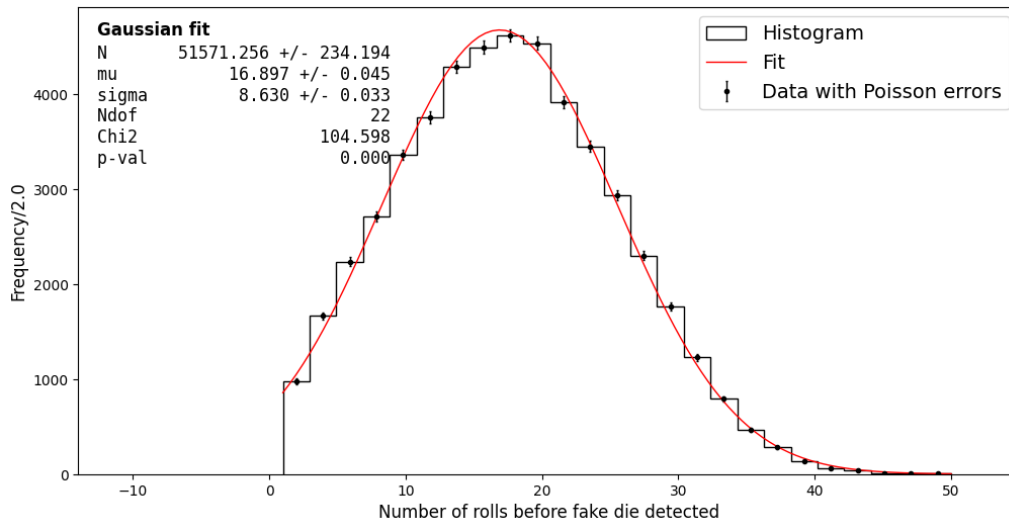


Figure 12: χ^2 Gaussian fit on top of fig. 11. χ^2 , N dof and p-values are displayed on top.

From the fit we obtain a mean value of the number of rolls needed before statistically determining that the box contains a fake die is $N = (16.89 \pm 0.05)$ rolls.

It is though worth mentioning that the found χ^2 from the fit is 5 times larger than $N \text{ dof}$.

Alternatively, one may trace p -values as a function of N to see if it makes sense to do a cut-off. Doing this, both with and without a fake die, we get:

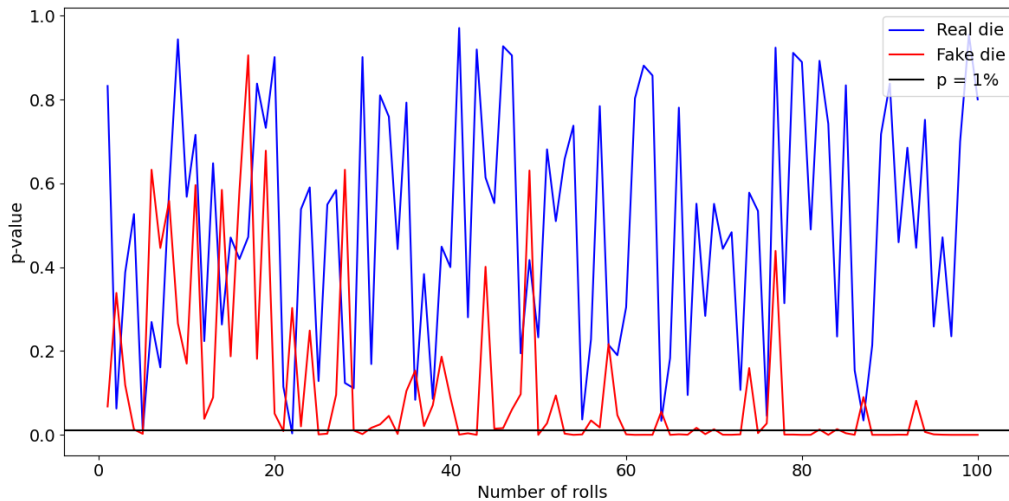


Figure 13: Trace of p vs. N for runs with and without a fake die.

From the figure, it is evident that even after once hitting the level of significance, the p -value fluctuates a lot. This means, we cannot be sure of whether our box contains a fake die just by finding the amount of rolls needed for p to be under said value. But it becomes more and more frequent after $N \geq 60$.

Problem V

5.1

5.1.1

Let us start by plotting the data:

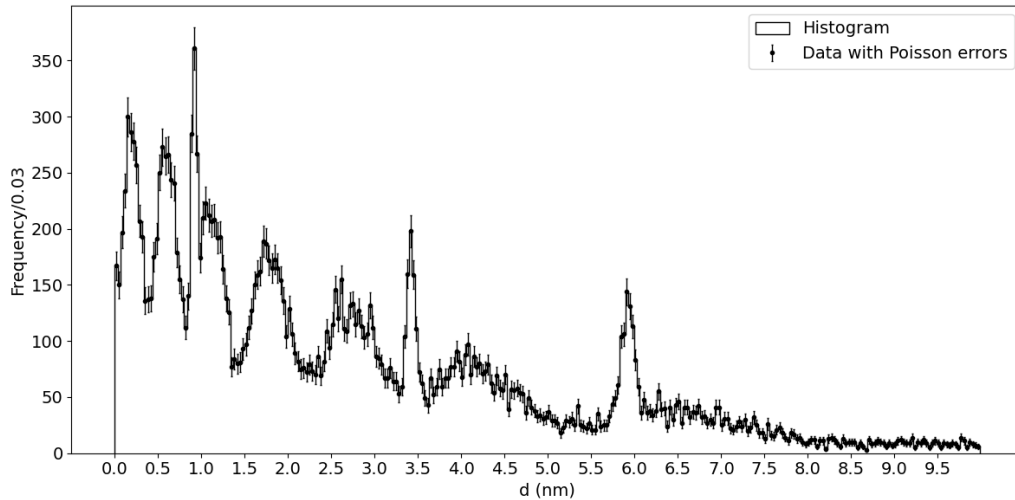


Figure 14: Distribution of distances with Poisson errors.

Now, let zoom in on $d \in [8, 10]$ and test to what extend the data is uniform.

To test, we do a Kolmogorov-Smirnov (KS) test. We thus have the null hypothesis, H_0 , that the distribution is uniform. We set a level of significance at 1%. To compare, we simulate a uniform distribution in the same range, and find the corresponding p-value.

Plotting the two together, we get the following:

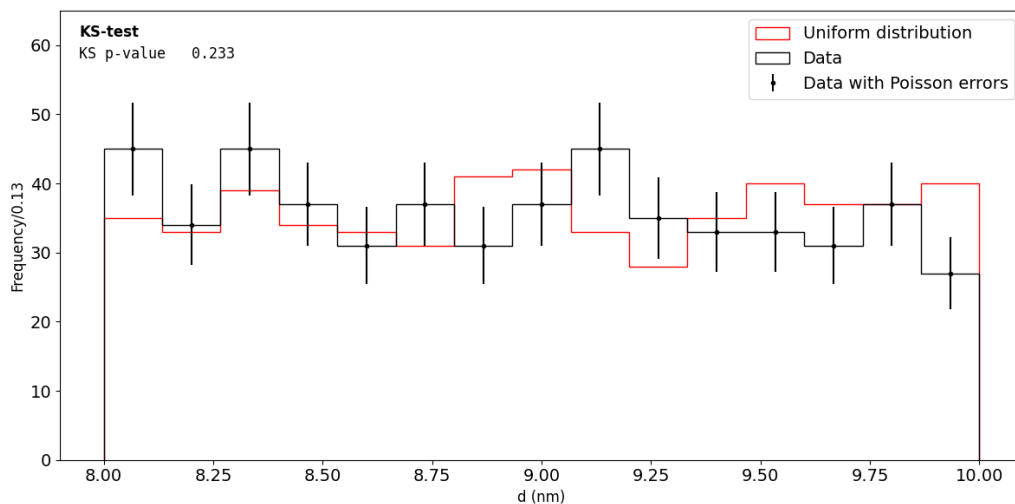


Figure 15: Distribution of distance for $d \in [8, 10]$ with a simulated uniform distribution drawn. The corresponding p-value for the KS test is displayed.

From the test we get at p-value of $p \approx 23.3\%$. We will thus accept our H_0 and conclude that the distribution is uniform.

5.1.2

To fit the three Gaussian peaks, we use a χ^2 -fit. Our approach is to cut the data around the approximate peak-values, and do a local fit. Looking at fig. 14, the total background looks approximately exponential. We thus use a combined gaussian and exponential fit function.

First looking at $d \in [0.8, 1.0]$, the first peak fit results in fig. 16:

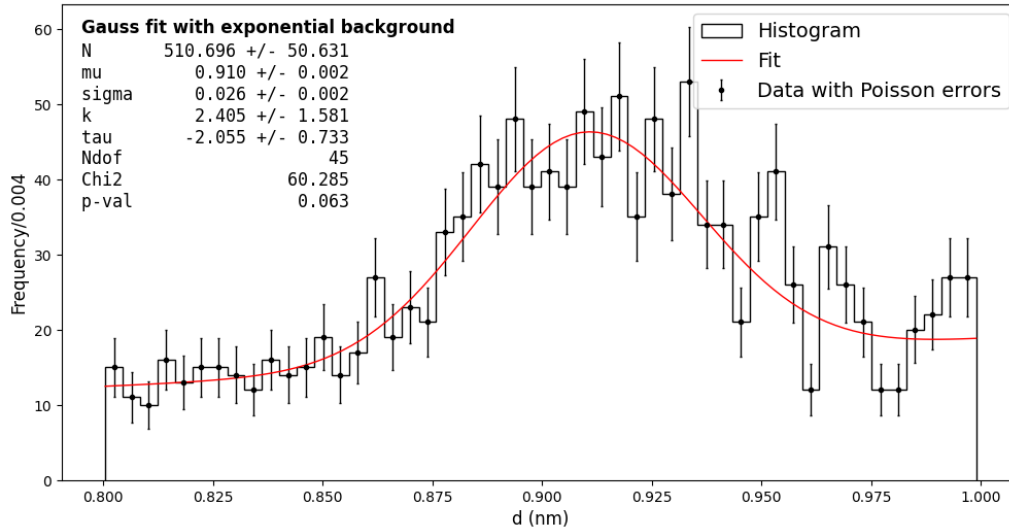


Figure 16: First peak fit for $d \approx 0.9$. The corresponding fit parameters are displayed on the figure.

Looking at $d \in [3.2, 3.6]$, we get fig. 17:

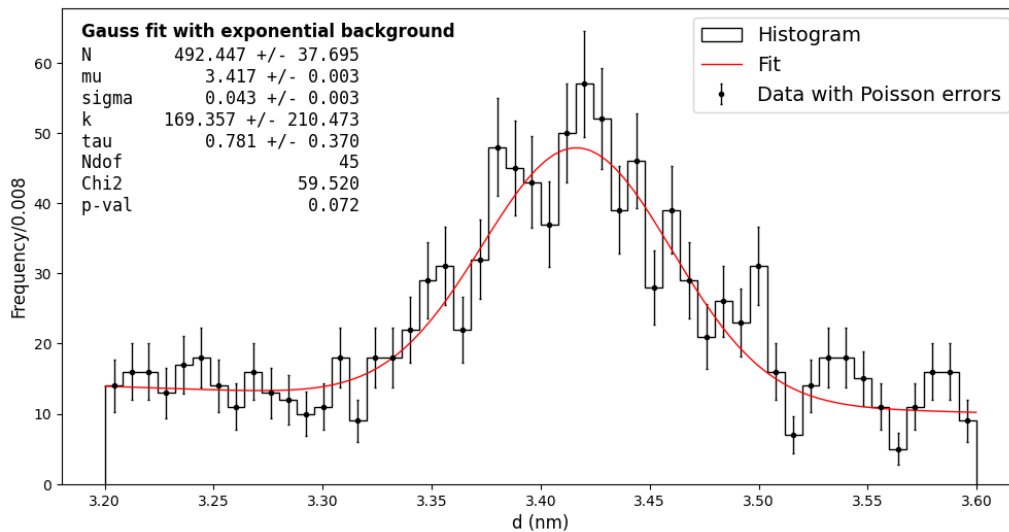


Figure 17: Second peak fit for $d \approx 3.4$. The corresponding fit parameters are displayed on the figure.

Looking at $d \in [5.7, 6.1]$, we get fig. 18:

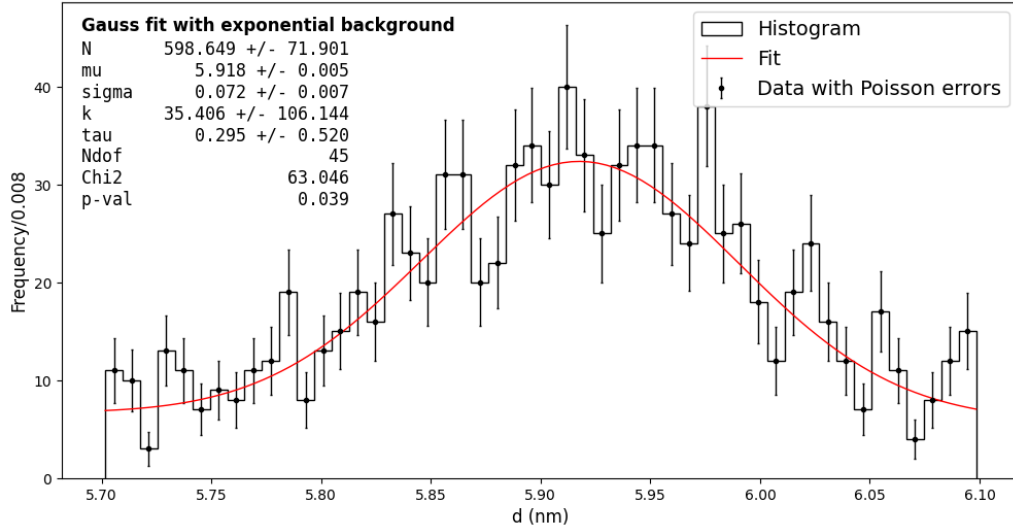


Figure 18: Third peak fit for $d \approx 5.9$. The corresponding fit parameters are displayed on the figure.

The relevant parameters are summarized in the following table:

Peak no.	Peak value (μ)	Intensity	$\chi^2/Ndof$	p
First	(0.910 ± 0.02) nm	(510 ± 50) points	60.28/45	0.06
Second	(3.417 ± 0.003) nm	(492 ± 37) points	59.52/45	0.07
Third	(5.918 ± 0.005) nm	(598 ± 71) points	63.04/45	0.04

Table 7: Relevant fit parameters from fig. 16, fig. 17 and fig. 18 summarized.

As can be seen in table 7, with a level of significance of 5% the fits are good, except for the last fit, which jumps just below this.

The initial guesses for the peak-centres match pretty good with the found values.

5.1.2

To test whether the peaks have the same intensity, we find the weighted mean and corresponding χ^2 , Ndof and p-value.

We get the following:

	Weighted intensity	$\chi^2/Ndof$	p
Combined results	(513 ± 28) points	1.72/2	0.42

Table 8: Combined intensities in a weighted mean with corresponding χ^2 , Ndof and p-values

With a null hypothesis, H_0 , stating that the intensities agree and a significant level of 1%, we accept that they have the same intensities, since the found p-value is much larger than level of significance.

5.1.3

To fit for the rest of the peaks, we choose some initial peak guesses and try to fit the data around them. The guesses are shown as dashed lines in the following:

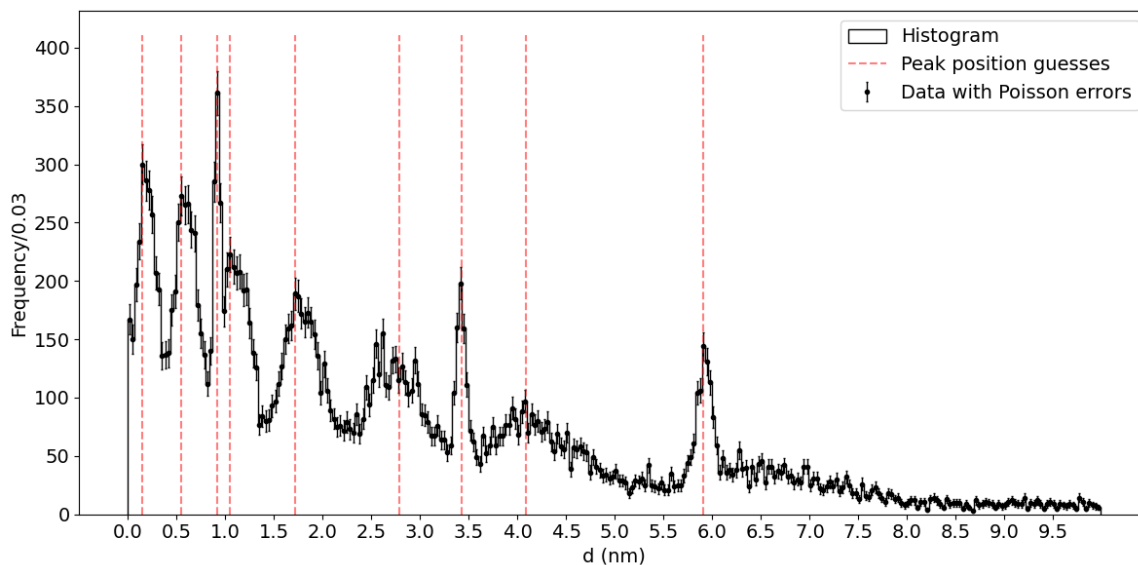


Figure 19: Distribution of distance with peak guesses drawn on top.

Our fit function consists of first fitting the entire exponential background and then adding each peak, using our start guesses for the positions of the peaks. Then minimizing χ^2 via Minuit.

The result can be seen below:

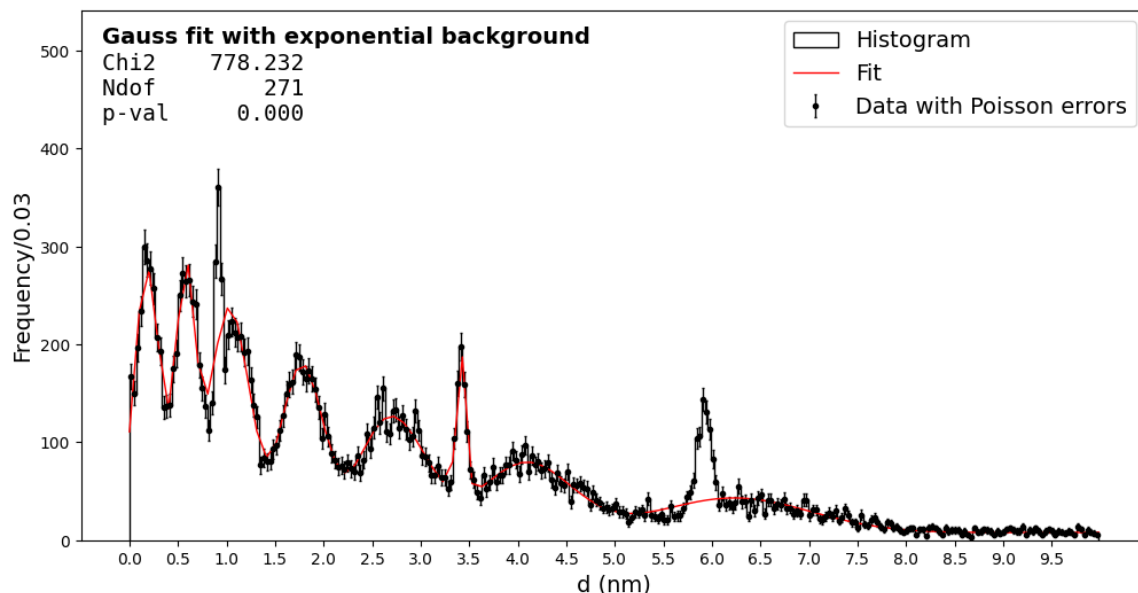


Figure 20: Total fit for all peaks in distance data. The fit χ^2 , Ndof and p-value is displayed.

As can be seen in fig. 20, our fit function cannot include all peaks. But it is very close! The found p-value is very(!) small and the χ^2 is pretty large compared to Ndof. To make a better fit, one may fit each peak by themselves, subtract them from the data, fitting the entire background, and then at last do a big fit. But I didn't have time for this...

5.2

5.2.1

To test to what extent the E and W values agree, we do a z-test. Our null hypothesis, H_0 , is that they do agree. We choose a significant level of 5%, and calculate the corresponding z- and p-values.

Before we do that though, we note that for 3. Red, 7. Sahure and 8. Nefer there are no West-values to compare to. By definition they agree then.

For the rest of the values, we get the following:

Pyramid	1. Meidum	2. Bent	4. Khufu	5. Khafre	6. Menk.
z-value	−1.77	−19.45	−2.12	0.00	−0.83
p-value	0.08	≈ 0	0.03	1.00	0.41

As can be seen, with a significant level of 5%, we must reject that the east and west measurements for 2. Bent and 4. Khufu agree. We accept our null hypothesis for the others, and state that the east and west measurements agree.

5.2.2

Let us combine the east and west values. For measurements with $|z| < 2$, we add a systematic uncertainty to both σ_E and σ_W until the z-value goes under 2. For the E,W pairs with no W measurements, we leave them alone and cite the weighted mean as just the east measurement.

Combining and adjusting we get the following:

Pyramid	1. Meidum	2. Bent	4. Khufu	5. Khafre	6. Menk.
z-value	−1.77	−1.99	1.96	0.00	−0.83
p-value	0.08	0.046	0.048	1.00	0.41
Comb. (arcmin)	(-19.4 ± 0.7)	(14.5 ± 1.4)	(-3.10 ± 0.15)	(6.00 ± 0.14)	(12.8 ± 0.8)

Table 9: Combined alignment values, along with the adjusted z- and p-values.

All the values are now in agreement.

5.2.3

To test if the pyramids shift linearly with time, we do a linear χ^2 fit.

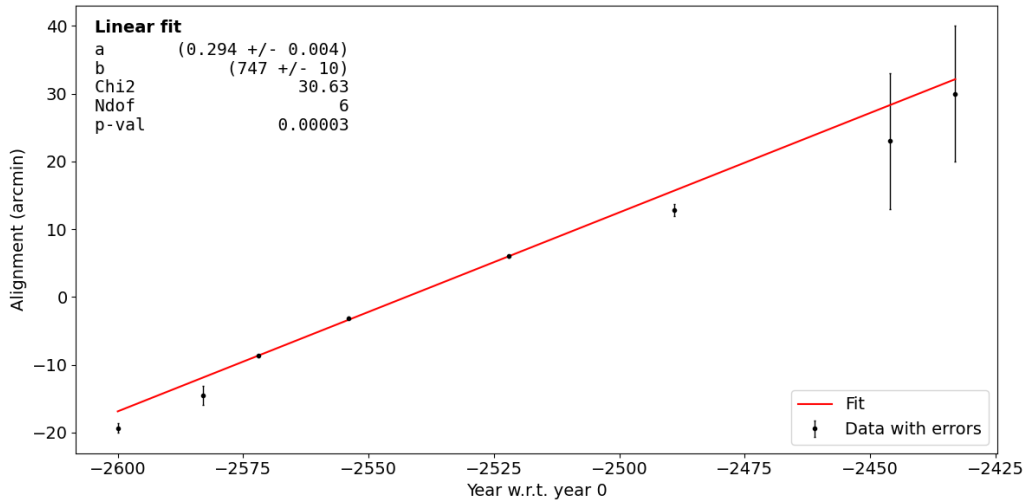


Figure 21: Linear fit to the combined alignment data. Fit parameters are displayed.

On the figure, we see a very small quoted p-value. This could be due to the very large uncertainties on the two rightmost data points. It could also indicate that the model is not linear. But, the amount of data points is very small.

The slope of the fit is:

$$a_{shift} = (0.294 \pm 0.004) \text{ arcmin/year} \quad (10)$$

5.2.4

To determine if the slope of the linear fit matches with the stated shift, we do a z-test. The null hypothesis, H_0 , is that they match. We choose a significance level of 5%.

The z-test gives a z-value of $z = 4.78$ and a p-value of $p = 0.0000018$. This is far below our chosen level of significance. We thus reject that they match.

5.2.5

Using $x = (y - b)/a$, with x denoting year, and using the found a and b values, we get the Khufu alignment year as (2545 ± 50) B.C. with the errors found through error propagation, of the above.